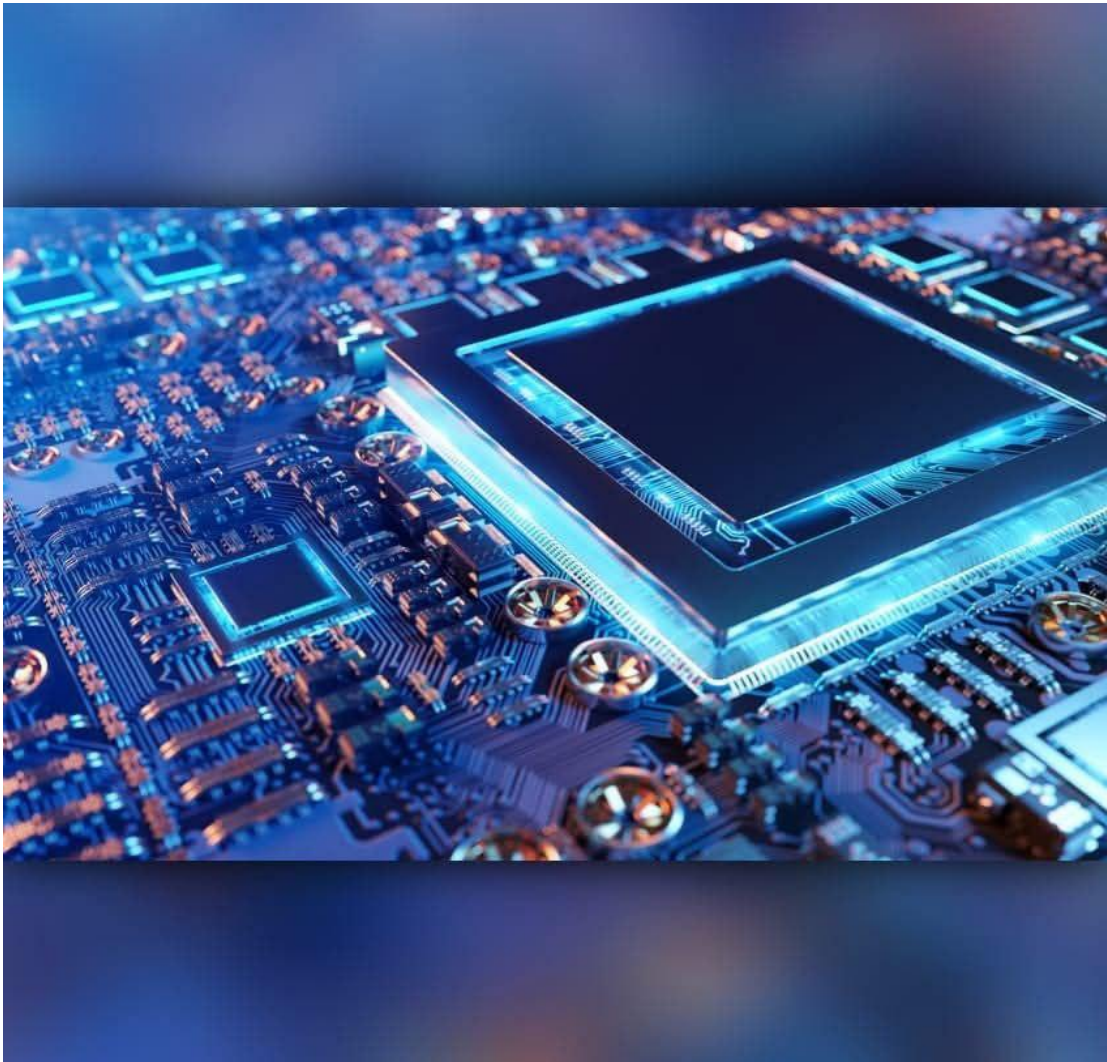


Digital Electronics Design
Assignment_Two_Extra

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Submitted to: Eng. Kareem Waseem

Question 1

Code



```
1  module encoding #(
2      parameter USE_GRAY = 1
3  )(
4      input [2:0] A,
5      output reg [6:0] B
6  );
7
8  generate
9      if (USE_GRAY) begin : gray_encoder
10         always @(*) begin
11             case (A)
12                 3'b000: B = 7'b00000000;
13                 3'b001: B = 7'b00000001;
14                 3'b010: B = 7'b00000011;
15                 3'b011: B = 7'b00000010;
16                 3'b100: B = 7'b00000110;
17                 3'b101: B = 7'b00000111;
18                 3'b110: B = 7'b00000101;
19                 3'b111: B = 7'b00000100;
20                 default: B = 7'b00000000;
21             endcase
22         end
23     end else begin : onehot_encoder
24         always @(*) begin
25             case (A)
26                 3'b000: B = 7'b00000001;
27                 3'b001: B = 7'b00000010;
28                 3'b010: B = 7'b00000100;
29                 3'b011: B = 7'b00001000;
30                 3'b100: B = 7'b00010000;
31                 3'b101: B = 7'b00100000;
32                 3'b110: B = 7'b01000000;
33                 3'b111: B = 7'b00000000;
34                 default: B = 7'b00000000;
35             endcase
36         end
37     end
38 endgenerate
39
40 endmodule
```

Tb_gray



```
1  module encoding_tb_gray;
2
3      reg [2:0] A;
4      wire [6:0] B;
5
6      encoding #(.USE_GRAY(1)) dut (
7          .A(A),
8          .B(B)
9      );
10     integer i;
11     initial begin
12
13         $monitor("A = %b , B = %b", A, B);
14
15         for (i = 0; i < 8; i = i + 1) begin
16             A = i;
17             #10;
18         end
19
20         $stop;
21     end
22
23 endmodule
```

Wave - Default

Variable	Value	Hex	Dec
A	3h7	00000003	3
B	7h04	00000007	7
i	32h00000008	00000000	0

Hex	Dec
00	00000000
01	00000001
03	00000003
02	00000002
06	00000006
07	00000007
05	00000005
04	00000004

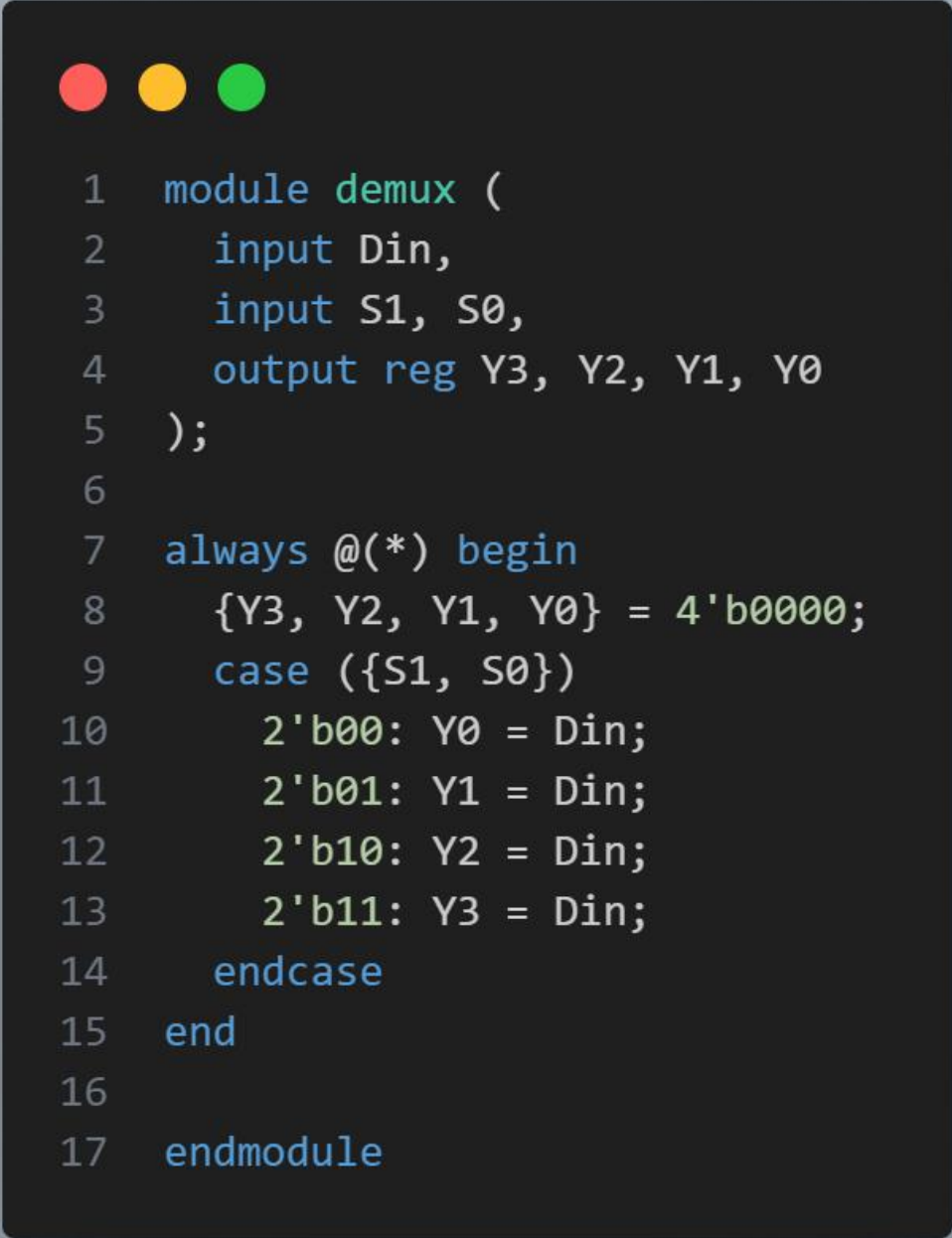
Tb_hot



```
1  module encoding_tb_onehot;
2
3      reg [2:0] A;
4      wire [6:0] B;
5
6      encoding #(.USE_GRAY(0)) dut (
7          .A(A),
8          .B(B)
9      );
10
11     integer i;
12     initial begin
13
14         $monitor("A = %b , B = %b", A, B);
15
16         for (i = 0; i < 8; i = i + 1) begin
17             A = i;
18             #10;
19         end
20
21         $stop;
22     end
23
24 endmodule
```


Question 2

Code



```
1  module demux (  
2      input Din,  
3      input S1, S0,  
4      output reg Y3, Y2, Y1, Y0  
5  );  
6  
7  always @(*) begin  
8      {Y3, Y2, Y1, Y0} = 4'b0000;  
9      case ({S1, S0})  
10         2'b00: Y0 = Din;  
11         2'b01: Y1 = Din;  
12         2'b10: Y2 = Din;  
13         2'b11: Y3 = Din;  
14     endcase  
15 end  
16  
17 endmodule
```



```
1 module demux_tb;
2
3   reg Din;
4   reg S1, S0;
5   wire Y3, Y2, Y1, Y0;
6   reg [3:0] expected;
7   integer i;
8
9   demux dut (.Din(Din), .S1(S1), .S0(S0), .Y3(Y3), .Y2(Y2), .Y1(Y1), .Y0(Y0));
10
11  initial begin
12      Din = 1;
13      for (i = 0; i < 4; i = i + 1) begin
14          {S1, S0} = i[1:0];
15          expected = 4'b0000;
16          expected[i] = Din;
17          #1;
18          if ({Y3,Y2,Y1,Y0} === expected)
19              $display("PASS: S=%b, Y=%b", {S1,S0}, {Y3,Y2,Y1,Y0});
20          else
21              $display("FAIL: S=%b, Y=%b (expected %b)", {S1,S0}, {Y3,Y2,Y1,Y0}, expected);
22          #9;
23      end
24      $stop;
25  end
26
27 endmodule
```



```

VSIM 17> run -all
# PASS: S=00, Y=0001
# PASS: S=01, Y=0010
# PASS: S=10, Y=0100
# PASS: S=11, Y=1000
# ** Note: $stop      : D:/QuestaSim Proj
#   Time: 40 ns   Iteration: 0   Instance
# Break in Module demux_tb at D:/Questa

```